

Employee Attrition Intelligence

Analysis Report

This report summarizes the completed employee attrition analysis and the key business insights identified from the IBM HR Employee Attrition dataset.

1. Executive Summary

The dataset contains 1,470 employees. 237 employees left, giving an overall attrition rate of 16.12%, while 1,233 employees stayed.

Metric	Value
Total employees	1,470
Employees who left	237
Employees who stayed	1,233
Overall attrition rate	16.12%

2. Department-wise Attrition

Sales has the highest department attrition rate at 20.63%, followed by Human Resources at 19.05%. Research & Development has the lowest rate at 13.84%.

Department	Attrition rate
Sales	20.63%
Human Resources	19.05%
Research & Development	13.84%

3. Overtime Analysis

Employees working overtime have a 30.53% attrition rate compared with 10.44% for employees who do not work overtime. This is a strong observed relationship, although it should not be interpreted as proof of causation.

Overtime	Attrition rate
Yes	30.53%
No	10.44%

4. Job Role and Salary Analysis

Attrition varies considerably across job roles. Sales Representatives are among the highest-risk observed roles, while several managerial and specialist roles show lower attrition. Average salary also varies by department and role, so compensation and career progression should be investigated alongside turnover.

5. Recommendations

- Reduce excessive overtime by monitoring workload and staffing.
- Prioritize retention programs in Sales and Human Resources.

- Investigate high-risk job roles using workload, satisfaction and career-growth indicators.
- Strengthen training, promotion pathways and internal mobility.
- Review compensation and incentives for high-turnover roles.
- Use predictive attrition modeling to identify elevated-risk employee profiles.

6. Conclusion

Employee attrition is not evenly distributed across the organization. Department, job role and overtime status show meaningful differences. The strongest observed pattern is the large gap between overtime and non-overtime attrition, while Sales and Human Resources also require attention. These findings support targeted, data-driven retention strategies.